

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B. Tech. Branch : Electronics and Computer Engineering/ Electronics and Computer Science

Subject Code & Name: BTESC304 Computer Architecture & Operating System Semester : III

Max Marks: 60

Date: 12/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Computer address bus is -	CO1	1
	a. Multidirectional b. Bidirectional c. Unidirectional d. None of the above		
2	Which of the following register can interact with the secondary storage?	CO1	1
	a. PC b. MAR c. MDR d. IR		
3	The addressing mode/s, which uses the PC instead of a general purpose register is _____	CO1	1
	a. Indexed with offset b. Relative c. Direct d. Both Indexed with offset and direct		
4	The time for the actual data transfer after receiving the request for data from secondary storage is referred to as the disk's?	CO2	1
	a. transfer time b. movement time c. access time d. data input time		
5	Which of the following computer memory is fastest?	CO2	1
	a. Register b. Hard disk c. RAM d. None of the above		
6	Which of the following is page fault?	CO2	1
	a. Page fault occurs when a program accesses a page of another program b. Page fault occurs when a program accesses a page in main memory c. Page fault occurs when there is an error in particular page d. Page fault occurs when a program accesses a page which is not present in main		

				memory		
7	The DMA differs from the interrupt mode by----				CO3	1
	a. The involvement of the processor for the operation	b. The method accessing the I/O devices	c. The amount of data transfer possible	d. nonintervention of CPU		
8	A set of microinstructions for a single machine instruction is called _____				CO3	1
	a. Program	b. Command	c. Micro program	d. Micro command		
9	What type of scheduling is round-robin scheduling?				CO4	1
	a. Linear data scheduling	b. Non-linear data scheduling	c. Preemptive scheduling	d. Non-preemptive scheduling		
10	Which of the following is not an operating system?				CO4	1
	a. DOS	b. LINUX	c. Windows	d. Oracle		
11	Which of the following is a condition that causes deadlock?				CO5	1
	a. Mutual exclusion	b. Hold and wait	c. Circular wait	d. All of these		
12	What is FIFO scheduling?				CO5	1
	a. First input-output scheduling	b. First in first out scheduling	c. Free input free output	d. All of the above		
Q. 2	Solve the following.					12
A)	State the difference between computer organization and architecture?				CO1	6
B)	Explain the different kinds of addressing modes with an example.				CO1	6
Q.3	Solve the following.					12
A)	Write a short note on 1) RAID Memory 2) Optical memory				CO2	6
B)	Explain in detail block diagram of memory management unit (MMU) and explain different functions performed by it?				CO2	6
Q. 4	Solve Any Two of the following.					12
A)	Differentiate between Interrupt driven I/O & Programmed I/O?				CO3	6
B)	Write a note on the Hardwired Control unit.				CO3	6

C)	Draw the microinstruction-sequencing organization of next-address field and explain it.	CO3	6
Q.5	Solve Any Two of the following.		12
A)	Define the Operating System. Describe the operating system's objectives and functions	CO4	6
B)	Write a short note on 1) Process and thread 2) Cooperating processes.	CO4	6
C)	State the scheduling criteria and explain any one scheduling algorithm in details with diagram?	CO4	6
Q. 6	Solve Any Two of the following.		12
A)	Explain the concept of semaphore? State its types	CO5	6
B)	Draw and explain concept of Process Synchronization?	CO5	6
C)	What is deadlock and explain different methods for handling Deadlocks?	CO5	6
*** End ***			